12.6.2

Note: 'Obs' and 'Ob' both mean observation
(I) apologize for the inconsistency)

	Obs1	Obs2	Obs3	Ob4
Ob1	0	0.3	0.4	0.7
Ob2	0.3	0	0.5	0.8
Ob3	0.4	0.5	0	0.45
Ob4	0.7	0.8	0.45	0

2a) Round 1:

Cluster 1 = {Obs1} Cluster 3 = {Obs3}
Cluster 2 = {Obs2} Cluster 4 = {Ob4}

Getting minimum dissimilarity among all clusters' pairwise dissimilarity values (non-zero):

$\min(0.3, 0.4, 0.7, 0.5, 0.8, 0.45) = 0.3$

So we first merge Ob1 & Ob2 into one cluster (this merge is at height 0.3):

Cluster 1 = {Obs1, Obs2}, Cluster 2 = {Obs3}, Cluster 3 = {Obs4}

Complete Linkage:

$$D(C_a, C_b) = \max_{i \in C_a, j \in C_b} d(i,j)$$

- $\max_{i \in C_1, j \in C_2} d(\{Obs1, Obs2\}, \{Obs3\})$

  $\max\ d(\{Obs1, Obs3\}, \{Obs2, Obs3\})$

  $\max(0.4, 0.5) = \underline{0.5}$

- $\max_{i \in C_1, j \in C_3} d(\{Obs1, Obs4\}, \{Obs2, Obs4\})$

  $\max(0.7, 0.8) = \underline{0.8}$

- $\max_{i \in C_2, j \in C_3} d(\{Obs3, Obs4\}) = \underline{0.45}$

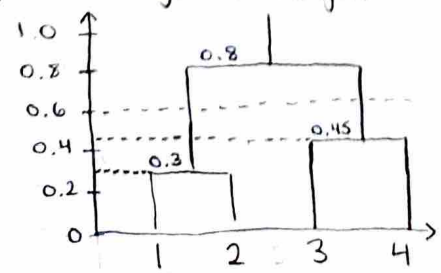
Minimum dissimilarity: $\min(0.5, 0.8, 0.45) = 0.45$ (next merge height)

so Obs3 & Obs4 merge into the other cluster: $\underbrace{\{Obs1, Obs2\}}_{C1}, \underbrace{\{Obs3, Obs4\}}_{C2}$

Complete Linkage: $\max d(\{Obs1, Obs3\}, \{Obs1, Obs4\}, \{Obs2, Obs3\}, \{Obs2, Obs4\})$

$= \max(0.4, 0.7, 0.5, 0.8) = 0.8$

Therefore, final merge at height 0.8 for $C1 = \{Ob1, Ob2, Ob3, Ob4\}$



Dendogram via complete Linkage

2c) If we split at height = 0.6, we get
C1 = {Obs1, Obs2}, C2 = {Obs3, Obs4}

2b) Dendogram via Single-Linkage (min instead of max)

Minimum dissimilarity among all pairwise observations:

$$\min(0.3, 0.4, 0.45, 0.5, 0.7, 0.8) = 0.3$$

Merge Ob1 and Ob2 at height 0.3 : $C1 = \{Obs1, Obs2\}, C2 = \{Obs3\}, C3 = \{Obs4\}$

Single-Linkage: $D(C_a, C_b) = \min_{i \in C_a, j \in C_b} d(i,j)$

- $\min d(\{Obs1, Obs3\}, \{Obs2, Obs3\})$ — C1 & C2
  $= \min(0.4, 0.5) = \underline{0.4}$
- $\min d(\{Obs1, Obs4\}, \{Obs2, Obs4\})$ — C1 & C3
  $= \min(0.7, 0.8) = \underline{0.7}$
- $\min d(\{Obs3, Obs4\}) = \underline{0.45}$ — C2 & C3

Minimum dissimilarity:

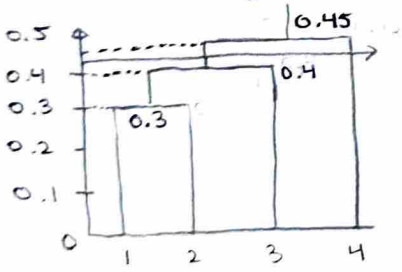
$$\min(0.4, 0.7, 0.45) = 0.4$$

So C1 and C2 merge at height 0.4 : $C1 = \{Obs1, Obs2, Obs3\}, C2 = \{Obs4\}$

Single Linkage:

$$\min d(\{Obs1, Obs4\}, \{Obs2, Obs4\}, \{Obs3, Obs4\})$$
$$= \min(0.7, 0.8, 0.45) = \underline{0.45}$$

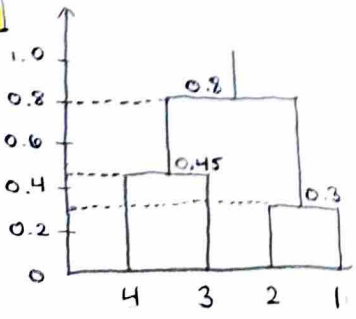
So we finally merge everything at height 0.45.



Dendogram via Single Linkage!

2d) At height = 0.425, we could slice and obtain: $C1 = \{Ob1, Obs2, Obs3\}, C2 = \{Obs4\}$

2e)



Leaf order now entirely different, and positions all swapped:
from → {1, 2, 3, 4}; to → {4, 3, 2, 1}
but cluster assignment / merging are still all the same, and the merge heights are also all the same, so this dendrogram is equivalent to 2a)'s.